

Why I Believe the REGAL Trial Will Fail and SLS Will Crash

Notes on galinpepimut-S, SELLAS Life Sciences (SLS), and the Phase 3 REGAL trial in AML

V2

by Franciszek Kessler

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I don't think REGAL is going to read out the way the bulls expect, and the stock's run-up into the data has gotten ahead of the evidence. Below is the case, roughly in the order I'd walk someone through it, with the sources at the end. I've also built a companion Excel [model](#) with my full valuation and TAM build-out; the numbers referenced in the valuation section below come from that model.

The base rate for this entire drug class is close to zero.

Start with the modality itself, before even getting to GPS specifically. Peptide vaccines that target tumor-associated antigens (TAAs) — proteins that are overexpressed in cancer but not mutated, like WT1 — have never produced a single FDA approval. Not one. An industry analysis tracking synthetic peptide vaccine candidates found that of more than 1,000 examined, 125 reached Phase I and 30 reached Phase II, and none has ever cleared Phase III to reach the market.

People sometimes point to sipuleucel-T (Provenge) as a counterexample, but that's actually a cell-based therapy — dendritic cells extracted from the patient and pulsed with a fusion protein outside the body — not a standalone injected peptide vaccine like GPS. Mechanistically, it's a different class of drug with a different manufacturing and regulatory path, so it doesn't really count against the pattern.

The literature is consistent on why this keeps happening: an immunosuppressive tumor microenvironment, tumor immune evasion, and tumor heterogeneity that lets low-antigen subclones survive selection. None of that is specific to a bad target choice — it's a property of the approach, and it doesn't go away because WT1 happens to be a well-validated target on paper.

Two prior examples make this concrete. Rindopepimut, a peptide vaccine for glioblastoma, had three positive Phase 2 trials, an FDA breakthrough designation, and a strong measured immune response — and then its placebo-controlled Phase 3 (ACT IV) was stopped early for futility. Tecemotide (Stimuvax), the MUC1 vaccine you and I already looked at, missed its Phase 3 primary endpoint in lung cancer despite years of buildup, and its own Phase 2 immune response rate was only around 21%. Both are close structural analogs to GPS: same broad mechanism, same encouraging early data, same outcome.

There's an even closer comp than either of those two, though: OCV-501, a WT1-targeting helper peptide vaccine developed by Otsuka—same antigen as GPS, not just the same broad drug class. Otsuka ran a proper randomized, double-blind, placebo-controlled Phase 2 trial of OCV-501 in 133 elderly AML patients in first complete remission, and the result was a clean miss: no significant difference in disease-free survival, the trial's primary endpoint, between the OCV-501 arm and the placebo arm. The only positive signal came from a post hoc subgroup analysis: patients who mounted a strong immune response (measured by specific IgG or interferon-gamma levels) had significantly longer overall survival than those on placebo. That's precisely the kind of result to be skeptical of. Post-hoc, non-prespecified subgroup analyses are a well-documented source of false-positive-looking findings, because you're searching after the fact for whichever slice of the data happens to look favorable. A same-antigen, randomized, placebo-controlled Phase 2 trial that missed its primary endpoint outright, salvaged only by a post-hoc immune-responder subgroup, is arguably the single most directly relevant precedent in this entire document. It's the same WT1 target GPS uses, a stronger baseline design than GPS's own foundational single-arm CR1 data, and a clean, disclosed primary-endpoint failure. Not a quiet fade into non-development. An actual negative trial with a paper published about it.

SELLAS has already run this exact experiment once, and it failed

This is the part that gets underweighted. SELLAS's other flagship drug, nelipepimut-S (**NeuVax**), is a HER2-targeting peptide vaccine using the same platform — a short peptide plus GM-CSF adjuvant — but aimed at breast cancer rather than AML. Its pivotal Phase 3 (the PRESENT trial, 758 patients, randomized, double-blind, placebo-controlled) was stopped early for futility. The independent monitoring committee reviewed the interim data and concluded that there was no realistic path to a positive result.

SELLAS kept the program alive afterward in narrower settings — a DCIS trial (VADIS), a terminated Phase 2b combination study, and some FDA discussions about a triple-negative breast cancer indication that didn't progress to an actual registration trial. So the pattern with this company's other big asset is: strong early data, a clean pivotal failure, then a retreat into smaller subpopulations rather than a full stop.

In terms of management specifically, Angelos Stergiou has run SELLAS since founding it in 2012. Before that, his C-suite and senior operational roles were at Genesis Life Sciences, PAION, Accentia Biopharmaceuticals, BioVest International, Analytica International, and Anavex Life Sciences, roughly across 2002–2009. I couldn't find a drug he led to FDA approval at any of these companies or at SELLAS — **there's no track record of a win to weigh against NeuVax's failure.**

The Anavex connection deserves a closer look, separate from the NeuVax point. Anavex's flagship drug, blarcamesine (ANAVEX2-73), has developed a genuinely rough public record in the years since its Rett syndrome Phase 2/3 trial (EXCELLENCE) missed its endpoints in January 2024. Its Alzheimer's Phase IIb/III trial has drawn a formal published complaint from outside researchers alleging inconsistent reporting of results over time, an undisclosed protocol and statistical analysis plan, and a co-primary endpoint (ADCS-ADL) that didn't reach significance even though the company's public messaging emphasized the endpoint that did. One independent report went further, describing a pattern of “spinning weak or failed clinical trial data into headline-friendly fluff.” I want to be careful here: Stergiou's time at Anavex was early, well before any of this unfolded, and I have no evidence he was personally involved in the conduct being criticized. But it's still a second company on his résumé, alongside SELLAS/NeuVax, where the flagship asset's public data story and the underlying trial results don't fully line up — and that's the kind of pattern worth just noticing rather than explaining away.

The biology is harder than the target rationale suggests

WT1 looks great on paper — it's overexpressed in nearly all AML cases and is mostly absent from normal adult tissue, which is why an NCI panel ranked it the top-priority cancer antigen back in 2009. But a good target on paper and a good vaccine target are two different things, and WT1 runs into the same wall most self-antigens do.

WT1 is a normal protein your body already makes, not a mutated, foreign-looking one. It's hard to make your body attack itself. The immune system spends your whole life deleting or muting T cells that react strongly to self-proteins, specifically to prevent autoimmunity — so a lot of the T cells you'd need for a powerful anti-WT1 response were never available to begin with. This is the textbook reason TAA

vaccines underperform, and it's why the field has been moving toward neoantigen vaccines (mutation-derived, genuinely foreign) instead. On top of that, a [2004 Journal of Clinical Investigation study](#) found a second, independent problem: some self/tumor peptides don't sit tightly enough in the MHC groove on the cell surface to hold a T cell's attention long enough to activate it — a stability issue, separate from tolerance. Think about it like this: the MHC molecule is a person holding out their hand, the peptide is the type of grip they use, the T cell receptor is another person deciding whether he or she wants to stop and talk to them. If the handshake is firm and steady, the two stay close enough to talk. If it's weak, contact is lost almost immediately. That's the point. Even if the T cell can recognize the peptide, it won't activate the peptide-MHC complex, because it's not stable enough to stay there.

There's also a straightforward escape route. AML is genetically messy — **lots of subclones, not one uniform population**. Put a WT1-vaccine-trained immune system under selection pressure, and the leukemic cells that already express less WT1, or that downregulate the MHC machinery needed to display it, have a survival advantage and can repopulate the disease without ever needing a new mutation. One study specifically looking at post-vaccination relapses didn't find WT1 loss or MHC downregulation driving them, which is mildly reassuring. Still, it's a small dataset with short follow-up, so I wouldn't lean on it too hard.

WT1 also sits inside the cell (it's a transcription factor), so ordinary antibodies can't reach it at all — that's exactly why this has to be a T-cell vaccine approach rather than an antibody drug. People have tried building TCR-mimic antibodies against the WT1-peptide-MHC complex instead. Still, none of that work has made it past the preclinical stage, which suggests this target has resisted more than one attempted solution, not just this one.

The Phase 2 data isn't the slam dunk it's presented as

Most of the bull case ultimately traces back to GPS's Phase 2 track record, so it's worth putting that data under the same scrutiny I'd apply to everything else in this thesis. It's weaker support than it's usually presented as, for three separate reasons.

First, the foundational trial (Maslak et al., Blood Advances, 2018; NCT01266083) was a single-arm, non-randomized trial. There was no concurrent control group — patients on GPS were compared against a historical benchmark drawn from older literature (a 20–25% expected 3-year OS rate), not against a contemporaneous group of similar patients followed at the same time, at the same centers, under the same standard of care. That's exactly the historical-control-bias problem I've flagged elsewhere in this document as a weakness of the REGAL comparator — except here it sits inside the primary piece of evidence used to justify running REGAL in the first place. A single-arm trial beating a historical control is real evidence of something. Still, it's a much weaker design than a randomized comparison, and it's the kind of design that gets a drug into a randomized Phase 3 rather than the kind that reliably predicts what that Phase 3 will show. It's possible the patients chosen for the Phase 2 were simply in a better shape than historically.

Second, and I think this is the more important point: that trial enrolled patients in CR1 — first complete remission — not CR2, which is the population REGAL is actually testing. CR1 patients haven't relapsed yet; they're a meaningfully healthier, better-prognosis group than CR2 patients who have already

relapsed once and gone through salvage therapy. Encouraging survival data in a first-remission population doesn't necessarily transfer to a second-remission population with a different, worse baseline prognosis. Citing the CR1 data as support for the CR2 pivotal trial is, in a biological sense, comparing results across two different diseases, even though both are labeled "AML in remission."

Third, the trial was tiny — 22 patients enrolled, and the widely cited 47% 3-year survival figure is based on just 19 evaluable patients. At that size, each patient's outcome is worth roughly 5 percentage points of the headline statistic. A single additional long-term survivor, or a single additional early relapse, would meaningfully move the number in either direction. The "median OS not reached, expected to exceed 67.6 months" framing sounds dramatic. Still, with a cohort this small, a median not yet reached is also just what you'd expect from too few events in too few patients — it doesn't have to reflect a large true effect size. The confidence interval around that number is almost certainly very wide, even though it isn't usually reported alongside the headline figure, just because of the small sample size.

There's also a separate, randomized CR2 Phase 2 comparison that's publicly cited (21.0 vs. 5.4 months OS) and that I've referred to earlier in this document as the strongest piece of supporting evidence, since it's randomized rather than single-arm. I want to flag some uncertainty about that data point directly. I have not been able to locate a clear, independently verifiable, peer-reviewed publication describing that specific randomized comparison with the same level of methodological detail as the CR1 paper above. It may exist and be less prominently indexed. Still, as things stand, the single most-cited piece of "randomized" support for GPS is, to my knowledge, less rigorously documented than the single-arm CR1 data. That's a strange asymmetry, since the single-arm data is the weaker design but the better-sourced one. I'd hold the randomized CR2 Phase 2 claim more loosely than it's typically presented, and I'd treat "a randomized supporting Phase 2 with a large effect size" (one of the genuine positives I list at the end of this document) as somewhat less certain than the way I originally framed it.

Three versions of the same idea: OCV-501, NeuVax, and GPS side by side

Now this is important. Between OCV-501 (Otsuka's WT1 helper-only vaccine), NeuVax (SELLAS' own HER2 killer-only vaccine), and GPS, you actually get about as close to a controlled comparison as this drug class offers. GPS shares its target antigen with OCV-501 and shares its developer and adjuvant philosophy with NeuVax. Laid out side by side, the comparison holds up.

	OCV-501	NeuVax	GPS
Target antigen	WT1	HER2	WT1
Developer	Otsuka	SELLAS (vía Galena)	SELLAS
T-cell mechanism	CD4 helper only	CD8 killer only	CD4 + CD8
HLA restriction	Class II (broader)	HLA-A2/A3	HLA-A*02:01 (2 of 4 peptides)
Adjuvant	Peptide alone	GM-CSF	Montanide + GM-CSF
Best available trial design	Randomized, double-blind, placebo-controlled (Phase 2, n=133)	Randomized, double-blind, placebo-controlled (Phase 3 PRESENT, n=758)	Single-arm, no concurrent control (CR1, n=22)
Reported immune response rate	100% DTH (Phase 1, n=9); ~24–38% post-hoc "responders" (Phase 2)	74% DTH (pooled Phase 1/2)	41% of the full CR1 population (best-documented); 64–80% in less rigorously documented later data

Clinical outcome	Missed the primary endpoint (DFS) in the randomized Phase 2	Stopped early for futility (Phase 3 PRESENT)	Not yet read out (Phase 3 REGAL ongoing)
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A few things jump out.

First, on the CD4/CD8 argument specifically: the claim that GPS is different from OCV-501 because it generates a CD8 killer T-cell response, not just a CD4 helper response, is mechanistically accurate. But NeuVax is a direct complication for that argument. NeuVax is pure CD8, no CD4 component at all, and it still failed its pivotal Phase 3. So “has a CD8 component” clearly isn't sufficient on its own to make one of these vaccines work — SELLAS' own other drug is a live demonstration of that. There's a real, legitimate immunology argument that combining CD4 help with CD8 killing should outperform either alone, since CD4 help is known to support CD8 memory formation. But that's a hypothesis GPS would be the first to actually validate in this family of drugs, not a pattern with confirming prior evidence — the record so far is one pure-CD4 failure and one pure-CD8 failure, not two failures that a combined approach has already shown it can fix.

Second, and I think this is the more damaging point: both OCV-501 and NeuVax **hit immune response rates at or above what GPS is currently boasting** — OCV-501 hit 100% DTH positivity in its Phase 1, NeuVax hit 74% DTH pooled across its early trials — and neither result translated into a clinical benefit once tested in a properly randomized trial. DTH is a cruder, less specific assay than the ELISPOT, tetramer, and antibody-titer methods behind GPS's own numbers, which if anything means the NeuVax and OCV-501 figures were probably easier to achieve, not harder. Immune response, as measured by any of the assays used across this entire family of drugs, has not been a reliable predictor of clinical benefit. That's not a criticism specific to GPS's own reporting; it's a pattern that shows up across every version of this approach anyone has tried.

Third, notice which of the three has the weakest supporting trial design. Both OCV-501 and NeuVax were tested at their most decisive stage in randomized, double-blind, placebo-controlled trials — the proper rigorous design. GPS's own best-documented data, by contrast, is the single-arm CR1 trial discussed in the previous chapter, with no concurrent control at all. The two drugs in this comparison that actually got the rigorous trial design both failed. GPS is the only of the three drugs that hasn't yet been tested that way at a pivotal scale — REGAL is that test.

Put together, what you have is two different configurations of the same broad approach (helper-only, killer-only), from two different companies, targeting two different antigens, both failing at the pivotal stage despite generating measurable immune response rates, in some cases very high. GPS is the third configuration — both arms combined — and REGAL is the first time anyone has tested that combination in a randomized, placebo-free, but controlled pivotal trial. That's a reasonable thing to be curious about. It is not a reason for confidence going in.

The trial itself has some structural soft spots

REGAL is randomizing 127 transplant-ineligible CR2 patients 1:1 to GPS or the investigator's choice of best available therapy (BAT), open-label, with a final analysis triggered at 80 pooled deaths. A few things about that design give me pause.

- 127 patients and 80 events is a small trial for a definitive Phase 3 — there's real room for a modest true effect to land on either side of significance just from chance.
- It's open-label, not blinded. That tends to introduce soft biases — follow-up frequency, threshold for starting rescue therapy — that generally inflate the treated arm's apparent benefit.
- The trial's historical benchmark (roughly 6 months expected survival for this population) may be stale. Venetoclax-based regimens and oral azacitidine both became standard of care in 2020, right as REGAL launched in February 2021 and enrolled through 2024. If the modern BAT arm is doing better than that old assumption, it dilutes GPS's apparent benefit even if GPS is doing exactly what it did in Phase 2. Against that: venetoclax-based therapy in the closest comparable setting (active relapsed/refractory AML) still shows only around 6.1 months median survival, so this risk is real but not conclusively proven.
- There's a statistical multiplicity risk worth watching for at readout. REGAL has secondary endpoints beyond overall survival — leukemia-free survival, time to AML progression. If the primary OS endpoint narrowly misses, watch for the press release framing to lean on whichever secondary endpoint looks best; that's a well-worn move when a primary endpoint disappoints, and it doesn't change what the trial was actually designed and powered to show.
- Even in a scenario where the trial succeeds, there's residual manufacturing risk between readout and launch. Peptide-plus-Montanide-adjuvant vaccines have historically had real-world batch consistency and supply issues in other programs; that's a separate, unresolved risk sitting between a positive trial result and a smooth commercial launch, and it doesn't show up in a topline hazard ratio.

I've broken the timing question — whether REGAL taking longer than planned is good news or bad news — out into its own section below, because I think it deserves a full treatment rather than a bullet point.

Why a delayed trial is not the bullish signal it looks like

A lot of the bull case for GPS leans on the fact that REGAL has taken longer than originally planned to reach its event threshold, on the theory that if patients are living longer, the drug must be working. I think this reasoning has the mechanism backward more often than not, and it's worth spelling out why in detail rather than just asserting it.

The more likely explanation: the original median OS assumption was wrong

REGAL is an event-driven trial — it needs 80 pooled deaths (across both arms combined) to trigger final analysis, regardless of how long that takes in calendar time. When a trial like this takes longer than planned, the arithmetic reason is always the same: fewer people are dying per month than the original

design assumed. The question is why. There are really only two possibilities, and they point in opposite directions:

- GPS patients specifically are living much longer than assumed, which is the bullish story; or
- The pooled population — which includes the control arm — is living longer than assumed for reasons that have nothing to do with GPS: a stale historical benchmark, a control arm that benefits from newer supportive-care drugs that didn't exist when the benchmark was set, or a trial population that is inherently healthier than the historical patients the benchmark was built from.

That second explanation, a miscalculated starting assumption rather than a working drug, has much stronger empirical support behind it as the typical cause. A systematic review of Phase III oncology trials from 2013 to 2023, published in ESMO Open, focused on trials that reached full enrollment but encountered statistical power issues. It found that 71.2% of those power failures were caused by a lower-than-expected event rate, and the dominant reason across the trials studied was that the control arm survived longer than the assumption used to power the study, not that the experimental drug outperformed expectations.

This isn't just a generic pattern. SELLAS has confirmed, on the record, that exactly this happened in REGAL. In a December 2022 [release](#), CEO Angelos Stergiou stated that a blinded review of the data conducted that fall showed patients living longer than expected, which "triggered modifications to the statistical analysis plan, in particular to reduce the number of events for the interim and final analyses." That's a direct, company-sourced admission that the pooled, blinded population (both arms combined, with no knowledge of which patients got GPS) was surviving past the original assumption, and the company's response was to lower the number of events required to reach a conclusion rather than treat the shift as ambiguous. Note carefully what this observation could and couldn't have told them. Since the review was blinded, it could only reflect the overall survival rate across both arms combined, not the GPS effect specifically. The company's own public framing at the time called this "very encouraging," but a blinded finding that patients are broadly living longer than assumed is neutral at best regarding whether GPS specifically works. It's equally consistent with the control arm benefiting from improved supportive care, which the systematic evidence above says is the more common explanation. A protocol change that reduces the number of events needed for a definitive readout is exactly the kind of amendment flagged as a red flag in the academic examples above, the everolimus, head-and-neck, and pancreatic cancer trials. The difference here is that in REGAL's case, the company itself confirmed the underlying reason for the change, rather than it having to be inferred after the fact.

The healthy-volunteer / trial-selection effect

There's a second, compounding reason trial populations tend to outlive historical benchmarks that has nothing to do with drug efficacy at all: patients who enroll in clinical trials are not a random sample of all patients with a given disease. Trials have eligibility criteria — adequate organ function, a minimum performance status, and no major uncontrolled comorbidities — that systematically screen out the sickest, frailest patients who would otherwise pull down a historical survival benchmark. A trial population, on both arms, will tend to outlive an unselected historical cohort simply because of who was allowed to enroll, independent of any treatment effect. Combine that with a genuine improvement in supportive care since the benchmark was set, and a pooled survival curve can end up looking

dramatically better than the historical assumption without either arm's actual treatment having anything to do with it.

Documented, named examples of exactly this pattern

This isn't a hypothetical concern — there are concrete, publicly documented trials where exactly this happened, and the sponsors themselves had to file protocol amendments acknowledging it:

- A Phase 3 adjuvant everolimus trial in kidney cancer (S0931) had to amend its protocol after actual median recurrence-free survival in the control arm came in at 6.6 years versus the 6-year assumption used at design — the control arm simply did better than planned, slowing event accrual and forcing the trial to add years of additional follow-up.
- A head-and-neck cancer trial testing radiation plus cetuximab after surgery saw a similar problem: its own protocol amendment attributes a multi-year delay directly to a lower-than-expected event rate, pushing the final analysis back by years.
- A pancreatic cancer adjuvant chemotherapy trial hit the same wall — a lower-than-expected event rate pushed the projected final analysis out to 2029 or later, and the trial's statisticians had to formally accept reduced statistical power rather than wait indefinitely for events to accumulate.

A secretly brilliant experimental drug was driving none of these three trials. In each case, the documented cause was a miscalculated baseline assumption about how the control arm — and the trial population generally — would behave, not a treatment effect. This is the same mechanism I think is most likely operating in REGAL: supportive care for AML (venetoclax combinations, oral azacitidine) improved substantially right around when REGAL launched and enrolled, and REGAL's population, like any trial population, was screened for eligibility in ways that make it healthier than the historical cohort its ~6-month benchmark was drawn from.

The pattern shows up repeatedly in small-cap biotech, too

Beyond the systematic academic evidence, there's a well-worn pattern: small-cap oncology stocks run up specifically when a trial is taking a long time, on the theory that the delay itself is good news — and it has an almost uniformly bad track record. Telik's (TELK) canfosfamide failed two separate Phase 3 ovarian cancer trials after years of anticipation, with one trial showing literally identical median progression-free survival (3.5 months) in both the drug arm and the control arm. GenVec's (GNVC) TNFerade is a particularly sharp illustration of the risk in reading an early positive-looking number too optimistically: a first interim analysis suggested roughly a 25% lower risk of death, but by the next scheduled look, months later, that had shrunk to a statistically meaningless 8% (hazard ratio 0.921), and the company scrapped the trial entirely. Novartis' (NVT) NOV-002 missed its Phase 3 endpoints in lung cancer despite encouraging earlier data. Cell Therapeutics' (CTIC) pixantrone showed a real effect that nonetheless wasn't enough to satisfy an FDA advisory panel, which voted unanimously against approval.

Compare all of that to what actually happens when a drug's effect is genuinely large: Roche's vemurafenib (Zelboraf) in melanoma, Onyx's sorafenib (Nexavar) in kidney cancer, Algeta's radium-223 (Alpharadin) in prostate cancer, and Johnson & Johnson's abiraterone (Zytiga) in prostate cancer were **all stopped early**, specifically because their independent monitoring committees saw a survival

benefit clear and large enough that continuing to randomize patients to the inferior control arm became both statistically unnecessary and ethically indefensible. Big, real drug effects tend to get caught quickly. **Trials that quietly run long, past their planned timeline, are empirically much more often explained by a miscalculated starting assumption than by a hidden winner.**

To be fair to the other side of this specific point: REGAL's January 2025 interim analysis did involve an independent committee reviewing unblinded data and confirming GPS exceeded the pre-specified futility threshold. That is a real, if modest, piece of positive information that none of the failure examples above had at a comparable stage. It doesn't override the base rate, but it means the delay isn't purely a bad sign here — it's just not the confirmation of success I think a lot of the market is treating it as.

The immune response data doesn't hold up as well as it's presented

The same skepticism applied to Stimuvax's 21% immune response rate is worth applying here. In the foundational GPS Phase 2 trial in CR1 AML ([Maslak et al., Blood Advances, 2018](#), n=22), only 14 of the 22 enrolled patients had any immune response data collected at all — 8 were simply never tested, some likely because they relapsed and dropped out early. Of the 14 tested, 9 were positive in at least one assay, which is where the commonly cited “64%” figure comes from. But 9 out of the full 22-patient population is 41%, not 64%, and the gap between those two numbers is exactly the kind of thing worth noticing.

The CD8+ killer T-cell response — arguably the more important one mechanistically — could only be measured in HLA-A*02:01-positive patients, which was 9 of the 22 enrolled. Most of the trial population couldn't even be evaluated on that measure. And when the authors checked whether immune responders actually did better than non-responders on survival, the difference didn't reach statistical significance (p=.11 and p=.08), and one patient with a confirmed T-cell response still relapsed anyway.

The more recent, better-sounding numbers — 64% in the randomized CR2 Phase 2, 80% in the REGAL interim — come from press releases and conference disclosures rather than a published paper with a disclosed sample size or per-assay breakdown. The trend has been toward more impressive headline numbers with less available methodology to verify them, which is the opposite of what I want to see.

Valuation: the market is already pricing this as a win

This is the part that ultimately matters most for the stock, independent of everything above. As of July 5, 2026, SLS is trading at \$14.98/share. With roughly 186 million shares outstanding, the market cap is about \$2.8 billion. Against roughly \$107 million of cash and no debt, that's an enterprise value of about \$2.7 billion for a company whose lead asset does not yet have a positive Phase 3 readout, let alone an FDA approval.

Sizing the actual commercial opportunity

Working through the addressable patient population for GPS in its lead indication: of roughly 20,000 new AML cases diagnosed annually in the U.S., perhaps 12,000 reach a first complete remission, of

whom a majority eventually relapse, and only a minority of those who relapse and receive salvage therapy go on to reach a second complete remission — one dataset put that CR2 rate at roughly 36% of relapsed patients. Removing the share who are transplant-eligible (REGAL's population is transplant-ineligible only) leaves a rough estimate of 1,000–1,300 new eligible U.S. patients per year, or perhaps 1,500–2,000 at steady state once you account for patients remaining on therapy for a period of time. Extending that to other developed markets outside Greater China (SELLAS licensed China rights separately) gives a rough global addressable population on the order of 3,000–4,500 patients at steady state.

At a plausible orphan-drug price in line with other targeted AML therapies (roughly \$150,000–\$200,000 per year, comparable to venetoclax or gilteritinib), that implies a peak, fully-penetrated total addressable market of roughly \$450 million to \$900 million annually — and realistic peak sales, assuming a more typical 30–50% share of that addressable population rather than full penetration, in the rough range of \$150 million to \$450 million per year.

A few things could push that realistic range lower still. GPS is licensed from Memorial Sloan Kettering, and I haven't been able to confirm exactly how many years of composition-of-matter protection remain once you layer that on top of standard orphan drug exclusivity (7 years from approval) — the actual monetizable window could be shorter than a simple peak-sales model assumes. Reimbursement is a separate friction: payers scrutinize a \$150,000–\$300,000-a-year therapy hardest exactly when the underlying survival benefit is modest and the population is narrow, which is GPS's situation if it clears the trial at all. And the competitive landscape isn't static — menin inhibitors like revumenib and other emerging AML maintenance approaches could carve into the same eligible population by the time GPS could plausibly launch, further capping the realistic addressable market from the demand side, not just the supply side.

What that means for the current price

An enterprise value of \$2.7 billion against a realistic peak sales estimate of \$150–450 million implies a multiple of roughly 6–18x realistic peak sales — for a single indication, pre-approval, ahead of a trial result I'd estimate has something like a 5% chance of even hitting statistical significance, before getting to the separate question of whether a narrow win translates into a commercially compelling, broadly reimbursed drug. Multiples like that are sometimes justified for a drug that has already been de-risked by an approval and is showing early launch trajectory. They are very hard to justify for an asset that is still waiting on its pivotal readout, for a drug class with a zero-for-history approval record, from a company whose only other pivotal trial on this exact platform failed. My read is that the current price is not really reflecting a 5% probability-weighted outcome — it's closer to what you'd expect if the market had already assumed the trial succeeds and the drug launches roughly on plan. That's the asymmetry I'd want anyone looking at this name to sit with: the downside case (trial disappoints, EV re-rates toward cash-plus-optionality) is a much larger percentage move than the upside case (trial succeeds, and the stock has to re-rate from a starting point that already assumed a lot of that success).

Worth stress-testing this with a genuinely generous bull case, too, because even the best-case math doesn't get you very far from here. Assume 4,500 patients (the high end of the global addressable population), a \$300,000 annual cost per patient (well above the \$150,000–\$200,000 comparable range

I used above), a 3% annual price increase to keep pace with inflation, and extremely fast adoption sustained over 15 years. That gets you to roughly \$2 billion to \$2.5 billion in peak annual sales — genuinely blockbuster territory. Layer on a generous 40% profit margin and discount the resulting cash flows back, and you land on an NPV of roughly \$5.4 billion, or about \$29 a share. That's a double from today's \$14.98 — which sounds attractive until you notice what had to go right to get there: extremely fast, near-total market adoption, a premium price nearly double what comparable AML drugs command, sustained pricing power for 15 straight years, and a 40% margin on top of all of it. That's a miracle commercial scenario, stacked on top of the separate miracle of the trial actually working in the first place (itself, by my estimate, only about a 5% probability event). Multiplying a generous bull-case commercial outcome by a roughly 1-in-20 probability of the underlying trial succeeding is not a combination that supports paying today's price for a double as the best realistic case.

Dilution, insider incentives, and thin coverage

A few structural points compound the valuation concern. Shares outstanding have grown 86.4% over the past year. Dilution on that scale, largely through warrant exercises, means any per-share value in this document is being measured against a moving denominator, and there's more room to come: SELLAS established an at-the-market (ATM) facility of up to \$150 million in 2026 that hadn't been used as of the most recent quarterly update, which is a real overhang sitting on top of the current share count regardless of trial outcome. Separately, CEO Angelos Stergiou received a grant of 1,065,000 restricted stock units on June 16, 2026, just weeks ahead of the anticipated readout, lifting his direct holdings to 1,759,183 shares. That's a compensation award, not an open-market sale, so it isn't the classic insider-dumping red flag, but a grant of that size timed immediately before a binary catalyst still says something about governance and incentive alignment. Finally, the stock is covered by only 2 to 4 analysts, which is thin even for a small-cap biotech. That means relatively little independent institutional scrutiny of the name, and it helps explain why retail sentiment (see below) can end up dominating the public narrative around the stock rather than being counterbalanced by sell-side coverage.

One fairness point on the other side: SELLAS also has a second pipeline asset, SLS009 (tambiciclib), a CDK9 inhibitor with positive Phase 2 signals in relapsed/refractory AML presented at ASH 2025 and now dosing in a newly diagnosed frontline AML study. Some portion of the current enterprise value plausibly reflects optionality on SLS009, not a pure bet on GPS/REGAL — I've priced the EV against GPS-only peak sales above, which somewhat overstates how purely GPS-dependent the stock is. That said, SLS009 is still early (Phase 2, unpartnered, in a crowded CDK9 field), and I don't think it comes close to closing the gap between today's price and a probability-weighted view of the REGAL outcome on its own.

A crowded short, too

One more thing here: the stock borrow rate on SLS has reportedly been running around 150% — an unusually high cost to borrow shares to short. That tells you demand to short the name is intense relative to the available float, which cuts two ways. It confirms there's a well-populated bear case already being expressed in the market (I'm clearly not alone in this view), but a hard-to-borrow stock with a lot of short interest is also exactly the setup that can produce a sharp, mechanical short squeeze

on any positive news, regardless of the news' actual durability or quality. Worth factoring into position sizing and timing, separate from the fundamental thesis itself.

Knowing who the fool at the poker table is

There's an old poker line: if you've been at the table for thirty minutes and don't know who the fool is, it's you. I'd apply that here. There are two dedicated retail communities built entirely around this stock, [r/SLS Stock](#) and [r/sellaslifesciences](#), and I think the mere existence of a fanbase around a single small-cap biotech ahead of a binary trial readout is itself a warning sign, independent of anything discussed in either forum. Reading through the posts, the focus is overwhelmingly on price action and bullish framing rather than on the underlying clinical or statistical facts — chart patterns, short-interest gossip, and speculation about what price the stock will hit, much more than serious engagement with trial design, historical base rates, or the mechanism. One post that stuck with me: a poster said that if SELLAS goes to \$40 billion, he's naming his son GPS. That's not a data point about the drug. It's a data point about the crowd holding the stock, and about how much of the current price is being driven by narrative and community enthusiasm rather than by a probability-weighted read of the actual evidence. A stock with an organized retail fanbase, a 150% borrow rate, and a valuation that already assumes success is a stock where it's worth asking, plainly, who's on the other side of your trade — and whether it's you.

Where I land

Weighing the class-wide zero-approval history, SELLAS' own prior Phase 3 failure on the same platform, the tolerance and antigen-escape problems specific to WT1, weaker-than-presented Phase 2 support (single-arm, wrong remission stage, small sample, and a randomized CR2 comparison I can't fully verify), a small open-label trial with a possibly stale control-arm assumption, the empirical pattern linking slow event accrual to underpowered or negative trials, immune-response data that's weaker than its headline framing suggests, and a valuation that looks like it already assumes success, against the genuine positives that remain (namely a weak historical comparator and an interim analysis that did clear its futility bar), I land at around a 5% probability of REGAL hitting its primary endpoint. Not a coin flip and not even close to one. A long shot. The genuine positives keep this from being a flat zero, but they don't come close to overriding a drug class with no approvals ever, a same-company same-platform prior failure, and a mechanism that fights self-tolerance on top of everything else. This is dramatically below what I think the current share price assumes.

Where the stock actually goes if I'm right

If REGAL fails, GPS is effectively worthless as a going concern, and the stock should trade down toward whatever is left: cash, plus whatever the market is willing to pay for the remaining pipeline. Cash and equivalents were \$107.1 million as of March 31, 2026. Against roughly 186 million shares outstanding, that's about \$0.58 per share in cash alone, and that \$0.58 is itself a ceiling, not a floor, since a failed pivotal trial would likely trigger cost cuts but also leaves a company that still needs to fund something going forward. Cash burn between now and any wind-down or pivot doesn't stop the day the data reads out. The other asset that would still exist is SLS009 (tambiciclib), the company's CDK9 inhibitor, which has shown positive Phase 2 signals in relapsed/refractory AML and is now dosing a newly

diagnosed frontline AML study. It's real, but it's early: Phase 2, unpartnered, competing in a crowded CDK9/cell-cycle inhibitor field, and the kind of risk-adjusted value the market typically assigns to an asset at that stage is modest relative to a name currently carrying a multi-billion-dollar enterprise value. Putting a generous, fully risk-adjusted value on SLS009 optionality on top of the cash gets you to something like \$0.80 to \$1.10 per share as a reasonable floor, which lines up closely with a straightforward \$1 ceiling estimate on the combined cash-plus-SLS009 downside case. Against a \$14.98 starting price, that's roughly 93 to 95% downside in the failure scenario, and at a 5% probability of success, that's the scenario that should be doing most of the work in anyone's expected-value calculation on this stock today.

None of this is a certainty. It's a case built on base rates, mechanism, and pattern-matching against a lot of prior peptide-vaccine history — exactly the kind of case that can be wrong in an individual instance even when the underlying reasoning is sound. But I think it's a stronger, better-evidenced case than “the trial is taking a long time, so it must be working,” and I think the current valuation leaves very little room for being wrong in the market's favor.

You can find my model for SLS on my GitHub: <https://github.com/nekkess>

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Note: valuation and TAM figures reference the author's companion Excel model (share price, share count, cash, EV, and peak-sales build-out), current as of July 5, 2026. Stock borrow rate is a market data point subject to change and should be re-verified at the time of reading.